

For each $e \in \mathbb{N}$, let $\varphi_e \in \mathbb{N} \rightarrow \mathbb{N}$ be the unary partial function computed by the RM with program $prog(e)$. So for all $x, y \in \mathbb{N}$: $\varphi_e(x) = y$ holds iff the computation of $prog(e)$ started with $R_0 = 0, R_1 = x$ and all other registers zeroed eventually halts with $R_0 = y$

Thus $e \mapsto \varphi_e$ defines an **onto** function from $\mathbb{N}$ to the collection of all computable partial functions from $\mathbb{N}$ to $\mathbb{N}$